

Seth Baird

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EDUCATION

- **Purdue University** West Lafayette, Indiana
Bachelor's in Computer Engineering Technology 08/2023 - 05/2027
Minor in Information Technology

PROJECTS

- **Modernized Superheterodyne FM Radio Receiver** 04/2025
 - Redesigned an analog superheterodyne FM radio with digital automated channel scan, presets, and a display.
 - Extrapolated the characteristics of the VCO using quartic regression over 56 data points with an R^2 of 0.9987.
 - The implementation used DSP to run frequency calculations and communicated with the tuning DAC over I2C.
- **Engine Monitoring Android App** 02/2024
 - Developed a data-driven driver companion app to aid in learning how to operate a manual transmission.
 - Aggregated engine sensor parameters (OBD-II PIDs) over MQTT and processed this data within the app.
- **Guitar Amplifier with Custom PCB** 12/2025
 - Engineered and fabricated a miniature guitar amplifier with a CAD-drawn case and a custom PCB in six weeks.
 - Simulated the circuit in Multisim and compared the frequency response curve against the one created in lab.
 - Directed a small team and utilized system diagrams, functional decomposition, Gantt, and critical path analysis.
- **Replica Arcade Controller** 11/2022
 - Deployed Rust code on a Raspberry Pi Pico in gadget mode to handle user inputs and transmit it back over USB.
 - Designed the HID hardware in OpenSCAD, allowing 4 variations of a part to be defined in a single file.

SKILLS

C | Java | Python | Embedded Systems | HAL | RTOS | Sensor Fusion | PCB Design | STM32 | FPGA

EXPERIENCE

- **Purdue Electric Racing** West Lafayette, IN
STM32 Embedded Software Team Member 09/2024 - 12/2025
 - **Register-level Programming** Developed the PWM HAL (hardware abstraction layer) to control onboard variable-speed devices like cooling fans. This made it possible to modulate the speed in two lines instead of 50.
 - **Lab Bench Testing** Leveraged oscilloscopes, logic analyzers, and specialized testing equipment such as CANable to verify the stability of safety-critical systems and communication channels.
 - **Embedded Architecture** Refined the vehicle's CAN layout by identifying a major communication bottleneck and ultimately relegating high-throughput motor commands to a separate network.
- **Wistron** Dallas, TX
NVIDIA Enterprise Technician 05/2024 - 08/2024
 - **Validation and Analysis** Utilized proprietary diagnostic tools to identify errors and discrepancies within motherboard subsystems such as power delivery, signal integrity, and FPGA before executing a repair.
 - **Team Workflow Optimization** Developed a Python script to automate thousands of lines of manual data entry into the company's antiquated database daily, saving my team 80 hours per week.
 - **Repair and Troubleshooting** Diagnosed, refurbished, and assembled NVIDIA HGX 8-GPU baseboards.
- **Jawa.gg** Remote
Verified Seller (PCs and Components) 10/2021 - 08/2023
 - **Top 10 Seller in Overall Performance** Recognized by CEO Amanda Stefan for driving Jawa's growth.
 - **Advanced Diagnostics** Analyzed schematics and BoardView files to diagnose deeper hardware issues, then performed board-level repairs using a BGA rework station and hand-soldering.

RELEVANT COURSEWORK

Advanced Embedded Systems | Concurrent Digital Systems | Data Structures & Algorithms | RF Comms.